Lecture plan · for the lecturer

Teaching optimisation at university: unit check

Optimization · University · Unit check

Time	Materials
40 min	Projector or board, and the lecture slides.

Learning goals

- Linear programming
- Integer programming
- Heuristics
- Modelling

Activity: Peer-instruction review

1. Vote (10 min): Show each question; students answer alone and commit to an answer.
2. Discuss (15 min): Where answers split, pairs argue for two minutes and vote again.
3. Explain (10 min): Work through the questions most students missed, showing the reasoning step by step.
4. Plan (5 min): Note which misconception to revisit next week and how.

Discussion

- Which misconception will you target first?
- Which lab will you add?
- How will you assess modelling?

Exit ticket

- One change to my optimisation course this term, and how I will know it worked.

Try it in your next lecture
Bring your open modelling brief and its rubric to your next department meeting for review.

Seminar sheet · for students

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Name _____ Class _____ Date _____

1 **For a bounded LP, an optimum occurs**

◯ At a vertex ◯ Only in the interior ◯ Nowhere

2 **Evaluate the corner.**

3x + 2y at (3, 1) = _____ .

Word bank: 11 · 12 · 9

3 **A constraint with slack at the optimum has a zero shadow price.**

◯ True ◯ False

4 **A maximisation relaxation gives 14. The integer optimum is**

◯ At most 14 ◯ At least 14 ◯ Exactly 14

5 **Compute acceptance.**

Δ = 2, T = 2: p = e^(_____).

Word bank: -1 · -2 · 1